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Multivariate analysis of the process of deep eutectic solvent pretreatment of lignocellulosic biomass

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ABSTRACT

In this study, multivariate analysis methods were used to reveal inner-relationship among key variables of deep eutectic solvents (DES) pretreatment of lignocellulosic biomass. Principal component analysis (PCA) and partial least square (PLS) were performed to analyze the effects of characterization of biomass material, physico-chemical properties of DES and pretreatment conditions on the efficiency of pretreatment in the choline chloride based DES pretreatment. The results revealed that temperature and severity of the reaction were the most important process parameters of the pretreatment. The physical and chemical parameters of DES related to hydrogen bonds were beneficial to get better delignification rate and recovery rate of glucan. This paper provided novel perspective for further understanding of DES pretreatment, designing DES, and optimizing process control of DES pretreatment.

1. Introduction

Recently, the research and development of renewable energy especially the bioenergy has been the focus of scientific research worldwide with the significant increasing energy demand (Nagarajan et al., 2020). Biofuels, such as bioethanol, emerged as most promising alternative to the non-renewable fossil energy, such as petroleum, coal and natural gas. Traditional fossil energy had obvious disadvantages such as environmental pollution and depletion of fossil resources (Daroch et al., 2013; Ha et al., 2020). For example, the global warming caused by carbon dioxide emission from the coal burning was the global urgent issues (Alexander et al., 2020). From the view of environmental protection and sustainable development, biofuel based on lignocellulosic biomass has been considered as the important contributors in energy field in the very near future (Kumar et al., 2020).

The production of biofuel in lignocellulosic biomass conversion process could be summarized as the following steps: pretreatment, enzymatic hydrolysis, fermentation and purification (Mosier et al., 2005). Pretreatment was an essential step, which could reduce the recalcitrance of biomass, break the connection between lignin and carbohydrates and increase the enzyme accessibility of cellulose (Xu et al., 2015). The development of economical and green pretreatment technology for the efficient fractionation of biomass was an urgent problem in the field of biomass energy research (Yu et al., 2016).Since Abbott reported in 2003, the new generation of green solvent deep eutectic solvents (DES), which had many similarities properties with ionic liquids, had attracted great attention in many different research fields (Abbott et al., 2003). DES had many advantages: low melting point, low cost, easy synthesis, high atomic economy, biodegradability, environmental friendliness, low toxicity and easy recycling (Elhamarnah et al., 2020). DES has showed good potential in breaking the recalcitrance property of biomass samples and disrupting the structure of the crystallinity (Kumar et al., 2019). In 2012, Francisco firstly reported that DES had significant differences in the solubility of lignin, cellulose and hemicellulose (Francisco et al., 2012). Previous reports suggested the effective use of DES in removal of lignin in different kinds of lignocellulosic biomass (Chen et al., 2018b; Dai et al., 2017; Hou et al., 2018; Procentese et al., 2015; Tan et al., 2018), which lead to more suitable substrate for the further bioethanol production process.

It should be noted that the emerging DES pretreatment was still in the early stage of research and the mechanism of DES pretreatment was still unclear. Not all types of DES were suitable for lignocellulosic biomass pretreatment. The properties of DES had great impacts on the

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pretreatment effect (Hou et al., 2018). Different types of DESs were prepared by mixing different kinds of hydrogen bond acceptor (HBA) and different kinds of hydrogen bond donor (HBD) with different molar ratios. Final properties of DES were affected by all characteristics of components in DES (Jablonsky et al., 2019). Moreover, DES pretreatment was a complex system with many variable parameters, such as parameters of DES properties and parameters of pretreatment conditions, thus, comprehensive analysis of the inner-relationships between all parameters in the whole process might be helpful for revealing the mechanism of DES pretreatment. Up to now, few paper published discussed this point, therefore, it was necessary to carry out multivariate analysis method (Xu et al., 2019) to get further understanding the interaction mechanism among the key parameters in whole DES pretreatment system. This method is mainly composed of principal component analysis (PCA) and partial least squares (PLS) (Ciccoritti et al., 2019). Until now, no multivariate analysis has been studied in the DES pretreatment.

In this paper, PCA and PLS analysis methods were performed on the choline chloride based DES pretreatment of different lignocellulosic biomass materials. The 54 important variables of the whole DES process were investigated. Correlations among variables were revealed for guiding future experimental designs to improve DES pretreatment of lignocellulosic biomass.

2. Methods

2.1. Materials

Different species of lignocellulosic biomass were studied for DES pretreatment process analysis in this paper. The datasets provided for multivariate analysis were based on agro-industrial food wastes including apple residues, potato peels, coffee silver skin, and brewer's spent grains (Procentese et al., 2018), switchgrass (Chen et al., 2018b), corncob (Procentese et al., 2015), sugar cane bagasse (Satlewal et al., 2019), waste lettuce leaves (Procentese et al., 2017).

The component analysis such as lignin content and carbohydrate content of raw material and pretreated biomass samples was conducted based on the National Renewable Energy Laboratory (NREL TP-510-42618) standard procedures (Sluiter et al., 2008). The results were shown in Table 1.

2.2. The properties of DES

Three different DES were studied for biomass pretreatment, which had same HBA (choline chloride) and different HBD including glycerol, ethylene glycol, and urea. The molar ratio of HBA to HBD was 1:2 for these three DES. The parameters such as interaction energy, void size, enthalpy of hydrogen bond formation, self-diffusion, coefficient, conductivity, density, surface tension, viscosity, solubility of glucose, solubility of carbohydrate, pH, molecular weight, boiling point, melting point, acentric factors, hydrogen bond acidity, hydrogen bond basicity, dipole or polarization, total energy binding (D'Agostino et al., 2011; Florindo et al., 2018; Hakkinen and Abbott, 2019; Lopez-Porfiri et al., 2016; Stefanovic et al., 2019; Xia et al., 2018) were discussed in this paper.

Table 1
Component of raw material.

Biomass	Glucan %	Xylan%	AIL%	ASL%
apple residues	31.3	6	30	2.9
potato peels	17.5	8	27	3
coffee silver skin	16.8	16.5	20	1.5
brewer's spent grains	21.2	12.2	16.5	2
waste lettuce leaves	28.02	22.14	15.01	3.1
corncob	31.5	22.1	13.7	3.2
switchgrass	31.57	19.11	20.13	0.23
sugar cane bagasse	43	24.8	24.3	1.6

2.3. DES pretreatment

DES pretreatments discussed in this work were performed under different pretreatment conditions. The pretreatment temperature were ranged from 60C to 150C, the pretreatment time were ranged from 1 h to 16 h, the solid to liquid ratio were ranged from 8 to 32, the pH of pretreatment system were ranged from 7.1–9.7. In order to get better evaluation of the severity of the pretreatment reaction, the modified severity factor (Hundt et al., 2014) based on time, temperature and pH was used in this analysis.

The evaluation parameters of DES pretreatment were solid recovery rate, recovery rate of glucan, recovery rate of xylan, delignification rate of lignin, which were evaluated by the variables as follows:

Biomass recovery (%) = (Pretreated biomass (g)/Original biomass (g)) × 100 %;

R-glucan (%) = (Biomass recovery × glucan content of pretreated biomass)/ glucan content of raw biomass × 100 %;

R-xylan (%) = (Biomass recovery × xylan content of pretreated biomass)/ xylan content of raw biomass × 100 %;

D-AIL (%) = 1–(Biomass recovery × AIL content of pretreated biomass)/ AIL content of raw biomass × 100 %;

D-ASL (%) = 1–(Biomass recovery × ASL content of pretreated biomass)/ ASL content of raw biomass × 100 %;

D-lignin (%) = 1–(Biomass recovery × lignin content of pretreated biomass)/lignin content of raw biomass × 100 %;

Wherein, Biomass recovery was the percentage of solid recovery. R-glucan and R-xylan were the recovery rate of glucan or xylan, respectively. D-AIL, D-ASL and D-lignin were the delignification rate of acid insoluble lignin, acid soluble lignin and total lignin of acid insoluble and acid soluble lignin, respectively.

2.4. Multivariate analysis methods

In this work, 54 important variables of DES pretreatment were conducted with PCA and PLS models, which were calculated using SIMCA-P software (Soft Independent modeling of Class Analogy, Umetrics).

3. Results and discussion

54 key parameters were divided into 6 groups as following, the parameters of HBD, the parameters of DES, DES pretreatment conditions, composition of raw biomass material, composition of pretreated biomass material and pretreatment evaluation parameters. The full name and the abbreviated code of 54 variables could be seen in Table 2.

3.1. Principal component analysis of the DES pretreatment of lignocellulosic biomass

PCA methods could be used to reveal and assess the relations and similarities among variables in the whole system (Sadalage et al., 2020). The PCA analysis results showed that it was valuable and reliable model with three principal components. The combined R square and Q square were 0.787 and 0.724, respectively, which meant that 78.7 % of total variability of the dataset and 72.4 % variations of all variables in this analyzed system could be explained well.

As shown in Fig. 1, there were three loading scatter plot. How the variables affected the entire dataset system could also be showed in plots according to the variables location and the distance between

Table 2
Variables in the multivariate analysis methods.

Calssification	Full name of variables	abbreviated code in models	PCA model	PLS 1 model	PLS 2 model	PLS 3 model	PLS 4 model
the parameters of HBD	Dipole moment	**MUP/HBD**	√	X	○	○	X
	Acentric factors	**omega/HBD**	√	X	○	○	X
	Acidity coefficient	**pKa/HBD**	√	X	○	○	X
	Molecular weight	**Mw/HBD**	√	X (T2)	○	○	X
	Refraction index	**RI/HBD**	√	X(T3)	○	○	X
	Melting point	**Tm/HBD**	√	X	○	○	X
	Boiling point	**Tb/HBD**	√	X (T1)	○	○	X
	Carbon number	**C/HBD**	√	X(T4)	○	○	X
	Number of hydroxyl groups	**OH/HBD**	√	X(T5)	○	○	X
	Number of amino groups	**NH2/HBD**	√	X	○	○	X
	Number of carbonyl groups	**C = O/HBD**	√	X	○	○	X
the parameters of DES	interaction energy	**interaction energy**	√	Y	X	○	X
	void size	**void size**	√	Y	X	○	X
	enthalpy of hydrogen bond formation	**ΔH**	√	Y	X	○	X
	self-diffusion coefficient of choline chloride (D-ch)	**D-ch**	√	Y	X	○	X
	self-diffusion coefficient of HBD	**D-HBD**	√	Y	X	○	X
	conductivity	**conductivity**	√	Y	X	○	X
	density	**density**	√	Y	X	○	X
	surface tension	**surface tension**	√	Y	X	○	X
	viscosity	**viscosity**	√	Y	X	○	X
	solubility of glucose	**S-glucose**	√	Y	X	○	X
	solubility of cellobiose	**S-cellubiose**	√	Y	X	○	X
	solubility of cellulose	**S-cellulose**	√	Y	X(T3)	○	X
	pH	**pH**	√	Y	X(T4)	○	X
	molecular weight	**Mw**	√	Y	X	○	X
	boiling point	**Tb**	√	Y	X	○	X
	melting point	**Tm**	√	Y	X	○	X
	acentric factor	**omega**	√	Y	X	○	X
	hydrogen bond acidity in Kamlet-Taft parameters	**alpha**	√	Y	X	○	X
	hydrogen bond basicity in Kamlet-Taft parameters	**beta**	√	Y	X	○	X
	dipole/polarization	**pi**	√	Y	X	○	X
	total energy binding	**EtN**	√	Y	X	○	X
DES pretreatment conditions	pH	**pH-R**	√	○	X(T5)	X	X
	highest temperature during pretreatment	**time-R**	√	○	X	X	X
	holding time at highest temperature	**T-R**	√	○	X(T1)	X(T4)	X(T1)
	the ratio of DES liquid to biomass samples	**LS ratio**	√	○	X	X	X
	combined severity parameter	**R factor**	√	○	X(T2)	X	X(T2)
	particle size	**size**	√	○	X	X	X
composition of raw biomass material	glucan content	**Glucan**	√	○	○	X(T5)	X
	xylan content	**Xylan**	√	○	○	X(T3)	X
	acid-insoluble lignin	**AIL**	√	○	○	X(T2)	X
	acid-soluble lignin	**ASL**	√	○	○	X	X
	acid-insoluble lignin and acid-soluble lignin	**Lignin**	√	○	○	X(T1)	X
composition of pretreated biomass material	glucan content	**P-Glucan**	√	○	○	Y	X
	xylan content	**P-Xylan**	√	○	○	Y	X
	acid-insoluble lignin	**P-AIL**	√	○	○	Y	X(T5)
	acid-soluble lignin	**P-ASL**	√	○	○	Y	X(T3)
	acid-insoluble lignin and acid-soluble lignin	**P-lignin**	√	○	○	Y	X(T4)
Pretreatment evaluation parameters	solid recovery rate	**Biomass recovery(%)**	√	○	Y	○	Y
	recovery rate of glucan	**R-glucan**	√	○	Y	○	Y
	recovery rate of xylan	**R-xylan**	√	○	Y	○	Y
	delignification rate of AIL	**D-AIL**	√	○	Y	○	Y
	delignification rate of ASL	**D-ASL**	√	○	Y	○	Y
	deligninfication of AIL and ASL	**D-Lignin**	√	○	Y	○	Y

√: This parameter was discussed in this model.
○: This parameter was not discussed in this model.
X: This parameter was set as X variables in this model.
Y: This parameter was set as Y variables in this model.
T1,T2,T3,T4,T5: the top first, second, third, fourth, and fifth most important variable in VIP analysis.

variable and coordinate origin. The farther the distance between the variable and origin, the stronger its influence would be. Variables closer to the origin had less impact on the entire system. For examples, the location of LS ratio was very close to the coordinate origin, which meant that solid to liquid ratio had less effect on the DES pretreatment compared with other condition variables. In the direction of p1, the most important variables which affect the process strongly were interaction energy, pi, EtN, beta. It was clear that the physicochemical

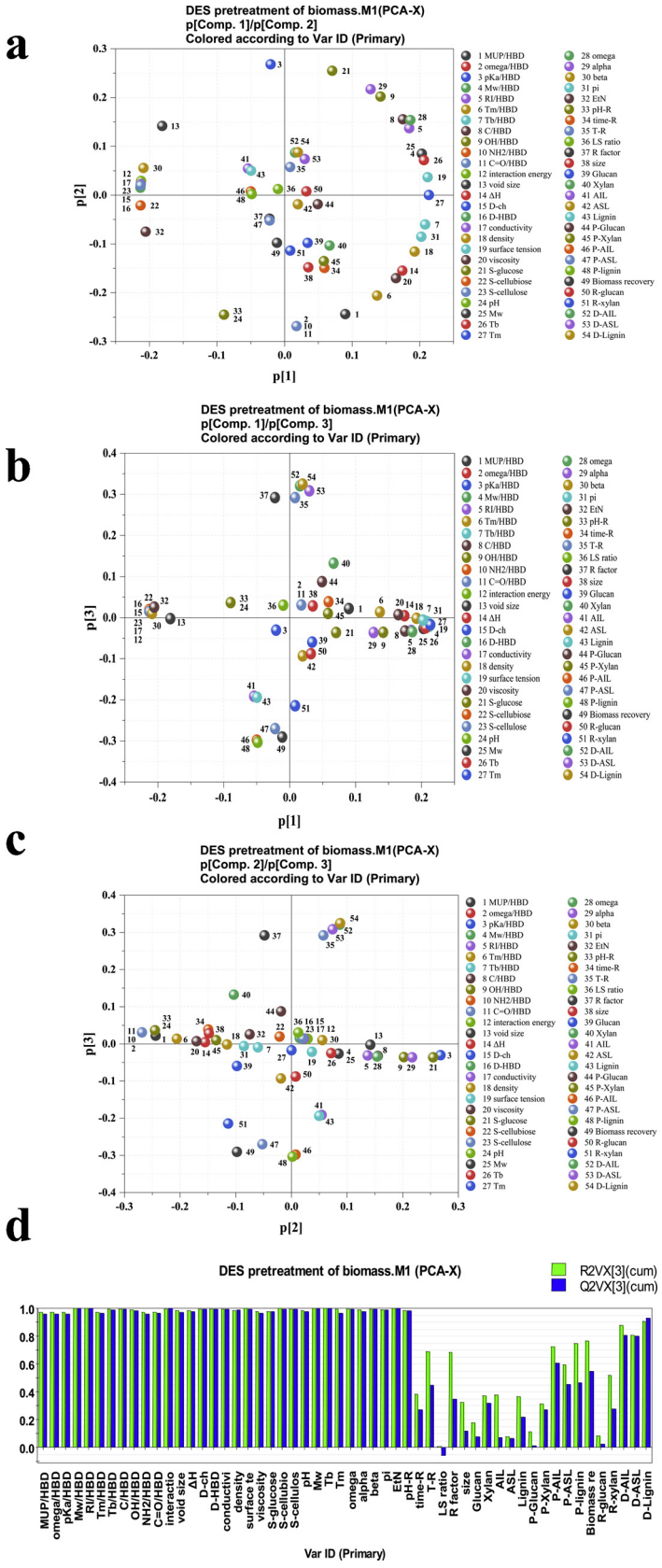


Fig. 1. Loading scatter plot for the PCA model of overall dataset.

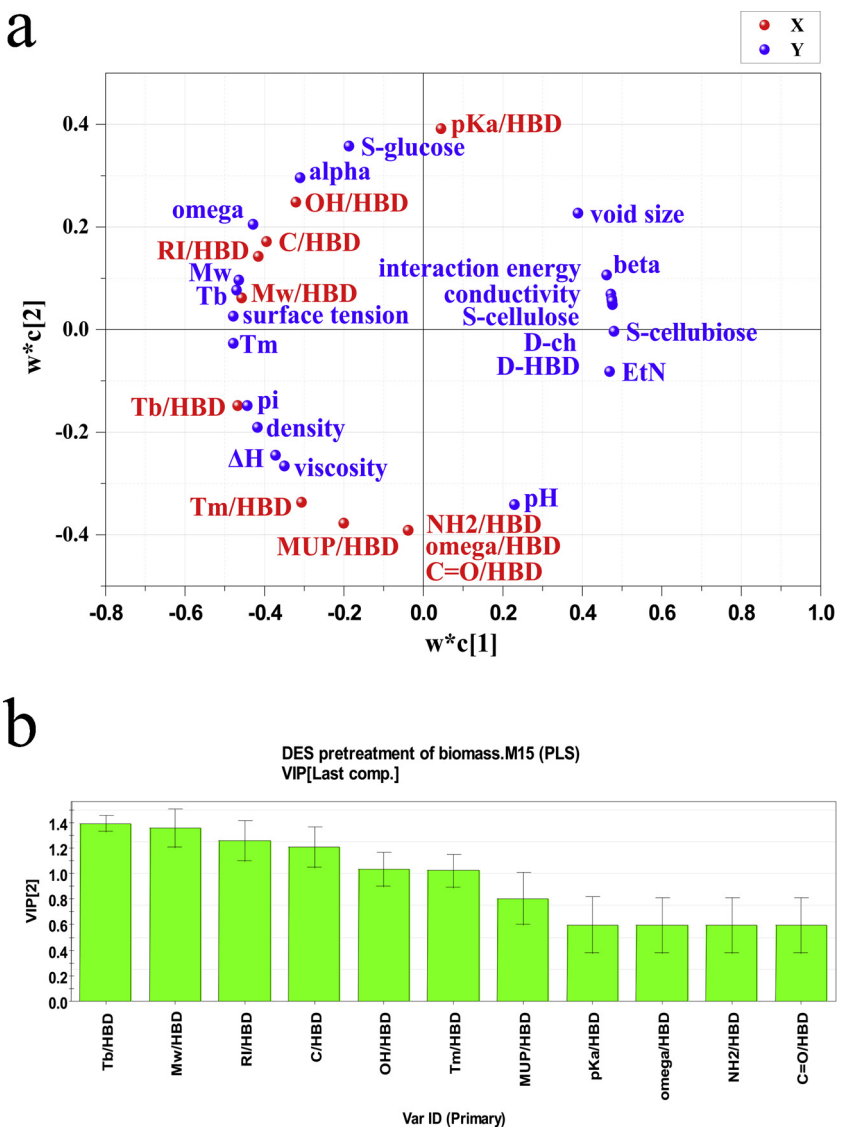


Fig. 2. PLS model 1. a) the loading scatter plot. c) variable important plot for the overall PLS model 1dataset.

properties of DES, especially energy, hydrogen bond strength and polarity had great influence on the effect of DES pretreatment(Florindo et al., 2018; Hakkinen et al., 2019). In the direction of p2, variables with heavy weight focused on physicochemical properties of hydrogen bond donors, especially those variables related to hydrogen bonding and polarity, such as acidity, number of hydroxyl groups, number of carbonyl groups, number of amino groups, acentric factors and dipole moment. These results were consistent with the discussion on the relationship between structure and function of DES in previous publications (Gilli et al., 2009; Hou et al., 2018). In the direction of p3, R factor was far away from coordinate origin, which meant the severity degree of reaction had strong influence on DES pretreatment.

The inner-relative relationships among variables were displayed in loading scatter plot. Variables near each other were positively correlated, and variables whose locations were opposite to each other were negatively correlated. As shown in Fig. 1a, the variables of DES pretreatment conditions, raw material composition, pretreated samples composition and pretreatment evaluation were located closely, which meant they had very strong relationship between each other. Specifically, for examples, as shown in Fig. 1b, variables of R factor, T-R, D-Lignin and D-ASL were grouped together closely. The reason was that compared to mild reaction condition, the increased temperature would lead to higher R factor (Goh et al., 2011), lignin would be removed easily in the more violent pretreatment(Alexander et al., 2020). Meanwhile, R factor and T-R were far away from Biomass recovery, for more degradation of carbohydrates would be get with the increased pretreatment temperature and more serious condition (Li et al., 2014).

The location relationship between the variables of physicochemical properties of DES and the structure of molecular could also be seen from this plot. As shown in Fig. 1a, alpha and OH/HBD were located closely, which revealed they were positively correlated. As the number of hydroxyl functional groups in the HBD molecule increased, the acidity of the DES hydrogen bond also increased (Hou et al., 2018). Meanwhile, it was obvious that the variable EtN was very far away from alpha and OH/HBD, which meant that total energy binding had strongly negative inner-relative relationship with them. The reason was easy to understand that if the hydrogen bond formed was stronger, the energy of the DES system would be lower and the DES would be more stable (Waite et al., 2018). Meanwhile, EtN was very far from boiling point and melting point of DES, which meant that they had serious negative correlation. It was because hydrogen bonding could affect Tb and Tm of DES. Tb and Tm were close to Mw/HBD for the reason values of melting and boiling point would increase with the increased molecular mass (Chen et al., 2018a).

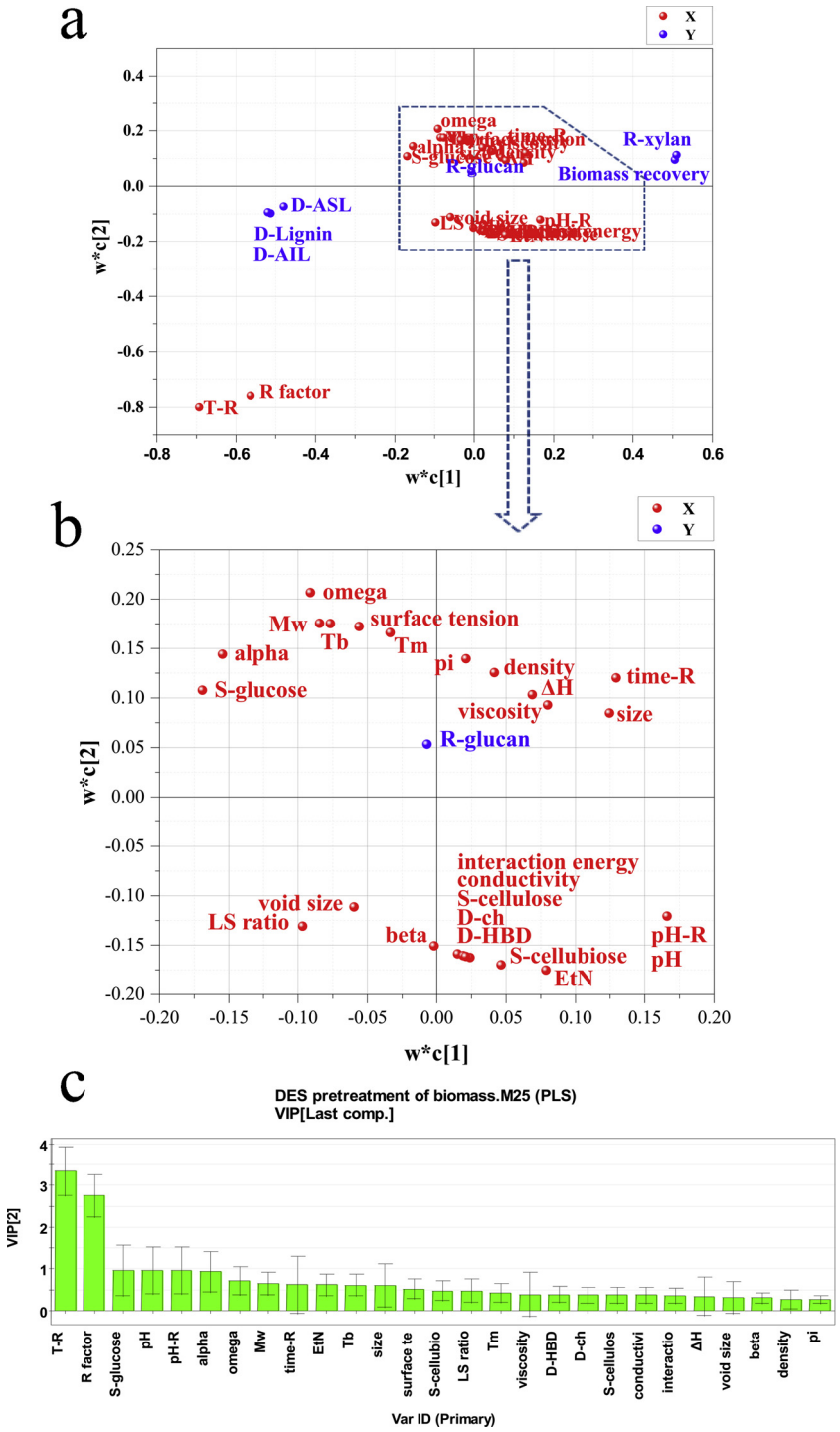


Fig. 3. PLS model 2. a) and b) the loading scatter plot. b) variable important plot for the overall PLS model 2 dataset.

3.2. Analysis of partial least square model 1 of the DES pretreatment of lignocellulosic biomass

There were 32 variables in PLS model 1. 11 variables of HBD were set as independent parameters, and 21 variables of DES were set as dependent parameters. The values of R2X, R2Y and Q2(cm) of this model were 100 %, which meant that this model was an excellent analysis model, which could explain the total information of 32 variables and have perfect predictive ability. As shown in Fig. 2a, in p1 direction, Tb/HBD, Mw/HBD, RI/HBD, C/HBD and OH/HBD were important variables and positively related with the polarity, hydrogen bonding properties, mass transfer properties and energy of DES. In p2 direction, pka/HBD, MUP/HBD and Omega were dominated. The importance order of effect of X parameter on all Y parameters could be seen in VIP plot in Fig. 2b.

The molecular weight and thermal stability such as boiling point of HBD had crucial influence on the physical properties of DES (Delgado-Mellado et al., 2018). As the molecular weight of HBD increased, the volume increased, the area of intermolecular interaction of DES increased, and the strength of the dispersion force of DES became greater (Chemat et al., 2016). The refractive index was related to the electronic polarizability of medium, so HBD with higher RI would be easier to be polarized (Craveiro et al., 2019). The polarity variable dipole moments and eccentric factors of HBD affected the value of polarity variable pi of

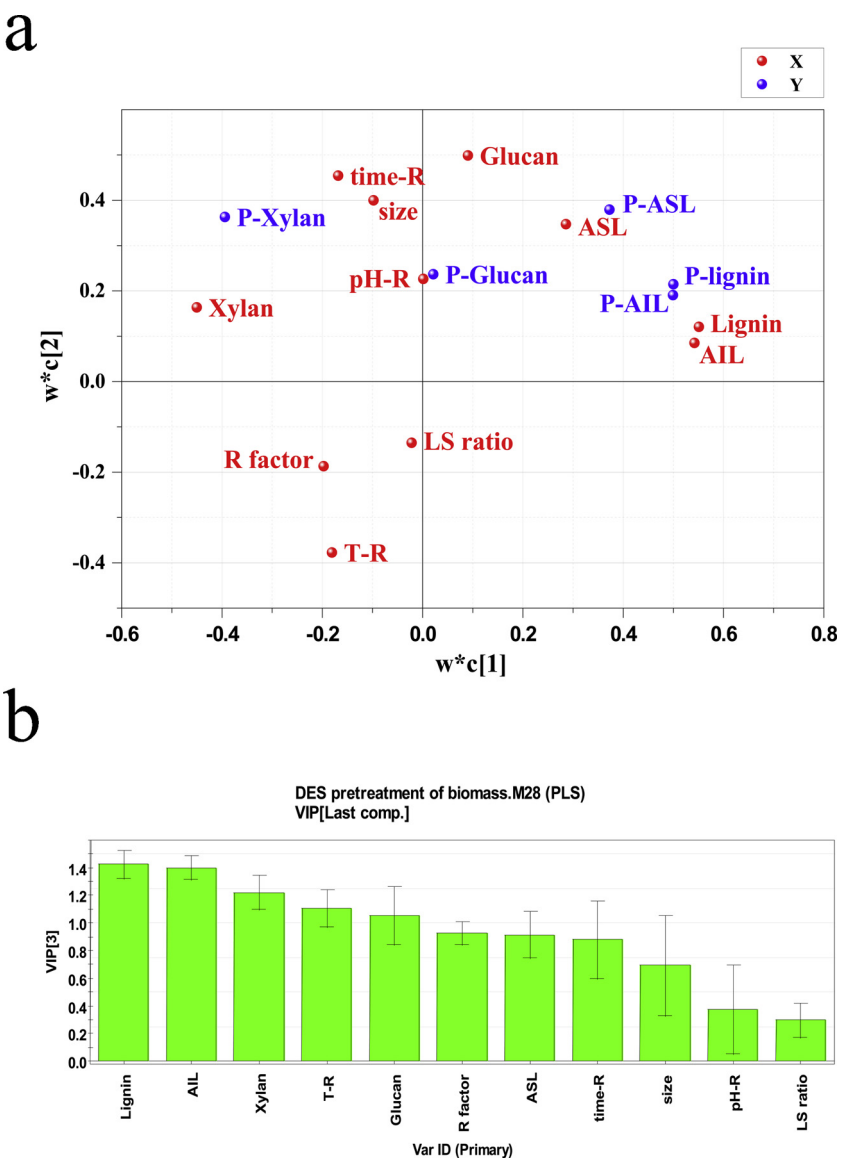


Fig. 4. PLS model 3. a) the loading scatter plot. b) variable important plot for the overall PLS model 3 dataset.

DES, which affected the formation of hydrogen bonds (Jablonsky et al., 2019). The characteristics of DES had very strong relationship with hydrogen bonding based on the HBD polarity when choline chloride was hydrogen bond receptor. The physicochemical properties of DES were dependent on the functional groups (Abbott et al., 2004), and hydroxyl group showed the most influence on this system hydroxyl among these three functional groups, hydroxyl, carbonyl, and amino groups.

3.3. Analysis of partial least square model 2 of the DES pretreatment of lignocellulosic biomass

To achieve better understanding of the effect of DES's properties and DES pretreatment conditions on the pretreatment results. There were 33 variables in this model, 27 variables were set as independent parameters, and 6 variables were set as dependent parameters. It showed that this model was a good PLS model, which could explain about 67 % information of the 33 variables of the analysis process, and have about 60 % predictive ability.

As shown in Fig. 3a and b, void size had significantly negative correlations with variables of density, surface tension, and viscosity, which were located in different quadrant. Viscosity, density and surface tension were macroscopic performance of microscopic interactions among molecules inside the DES system (Sanchez et al., 2019). The smaller the void size of DES system, the greater the liquid density and the higher the liquid viscosity would have (Hammond et al., 2016). The high viscosity of DES limited the solubility of carbohydrates (Hakkinen and Abbott, 2019) and viscosity affected mass transfer in the reaction process.

As shown in Fig. 3c, it was very obvious that the most important variables were T-R and R factor from the perspective of influencing the DES pretreatment efficiency. Pretreatment temperature had a stronger effect compared to other conditions variables such as holding time at highest temperature, solid to liquid ratio, pH, the particle size of raw biomass material. R factor reflected the severity of reaction was directly related to the efficiency of pretreatment (Goh et al., 2011). More severe reaction conditions caused by high temperature could lead to high value of delignification rate and carbohydrate degradation rate (Kumar et al., 2020).

3.4. Analysis of partial least square model 3 of the DES pretreatment of lignocellulosic biomass

PLS model 3 was to quantify the effect of pretreatment conditions

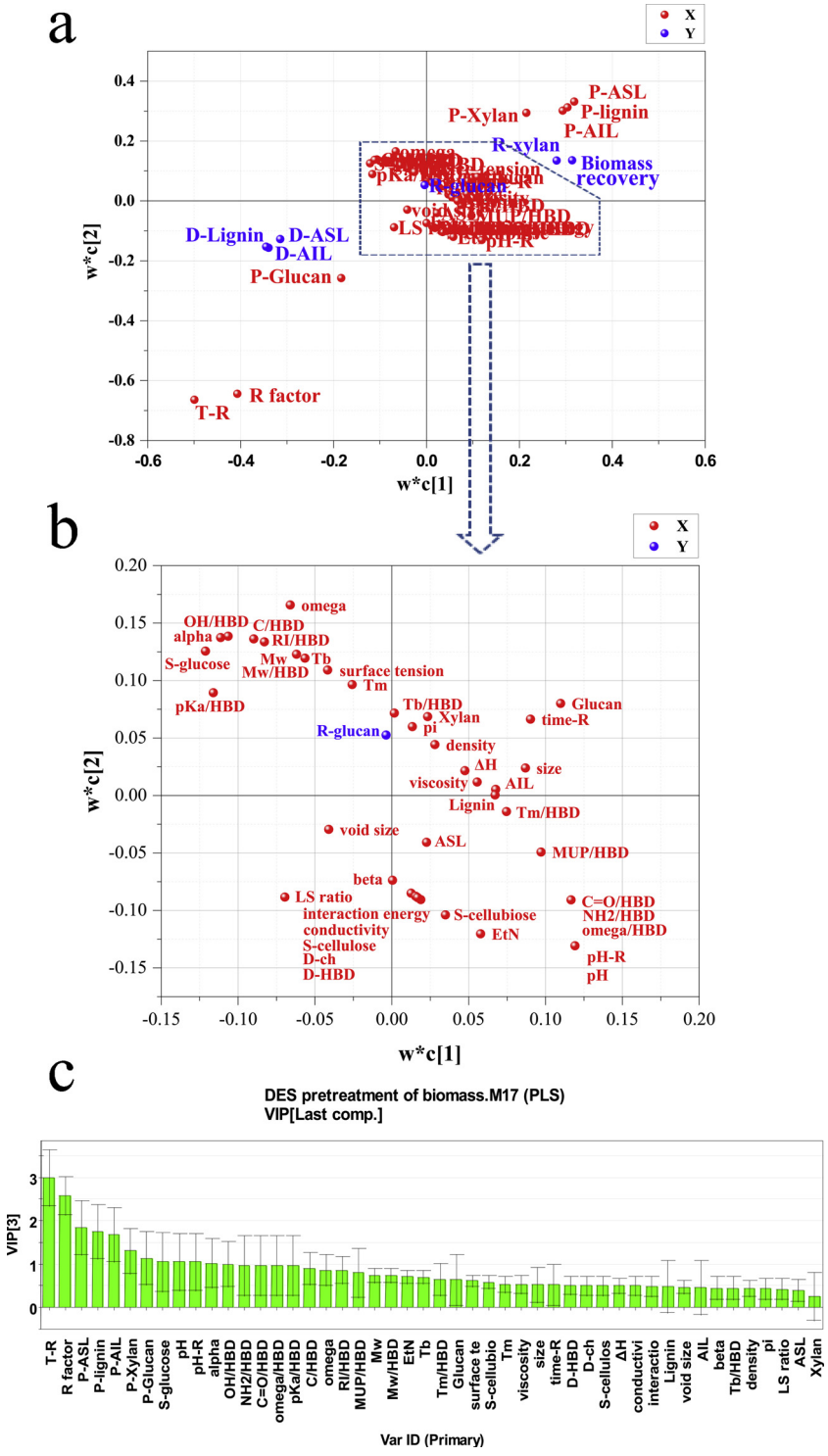


Fig. 5. PLS model 4. a) and b) the loading scatter plot. b) variable important plot for the overall PLS model 4 datasets.

and raw biomass material on the composition of pretreated biomass samples. There were 11 independent parameters and 5 dependent parameters. This PLS model could explain about 66.4 % information of X variables, 82.6 % of Y variables and have predictive ability of 81 %.

Fig. 4 showed that lignin content of raw material played the most important role on the composition of pretreated samples compared to other ingredients such as carbohydrate. The reason for that was lignin was partially responsible for the rigid structure and nature recalcitrance of lignocellulosic biomass in the biofuel conversion (Azar et al., 2020).

3.5. Analysis of partial least square model 4 of the DES pretreatment of lignocellulosic biomass

To investigate the impact of all the variables on the final pretreatment results, the variables of all six groups were selected for the PLS model 4 analysis. There were 48 X variables and 6 Y variables. This model could explain about 77 % information of the X variables, 68 % of Y variables, and have 65 % predictive capability. Kamlet-Taft fluorescent solvent discoloration probe has usually been studied the polarity of DES(Salehi et al., 2019). As shown in Fig. 5a and b, the DES of the

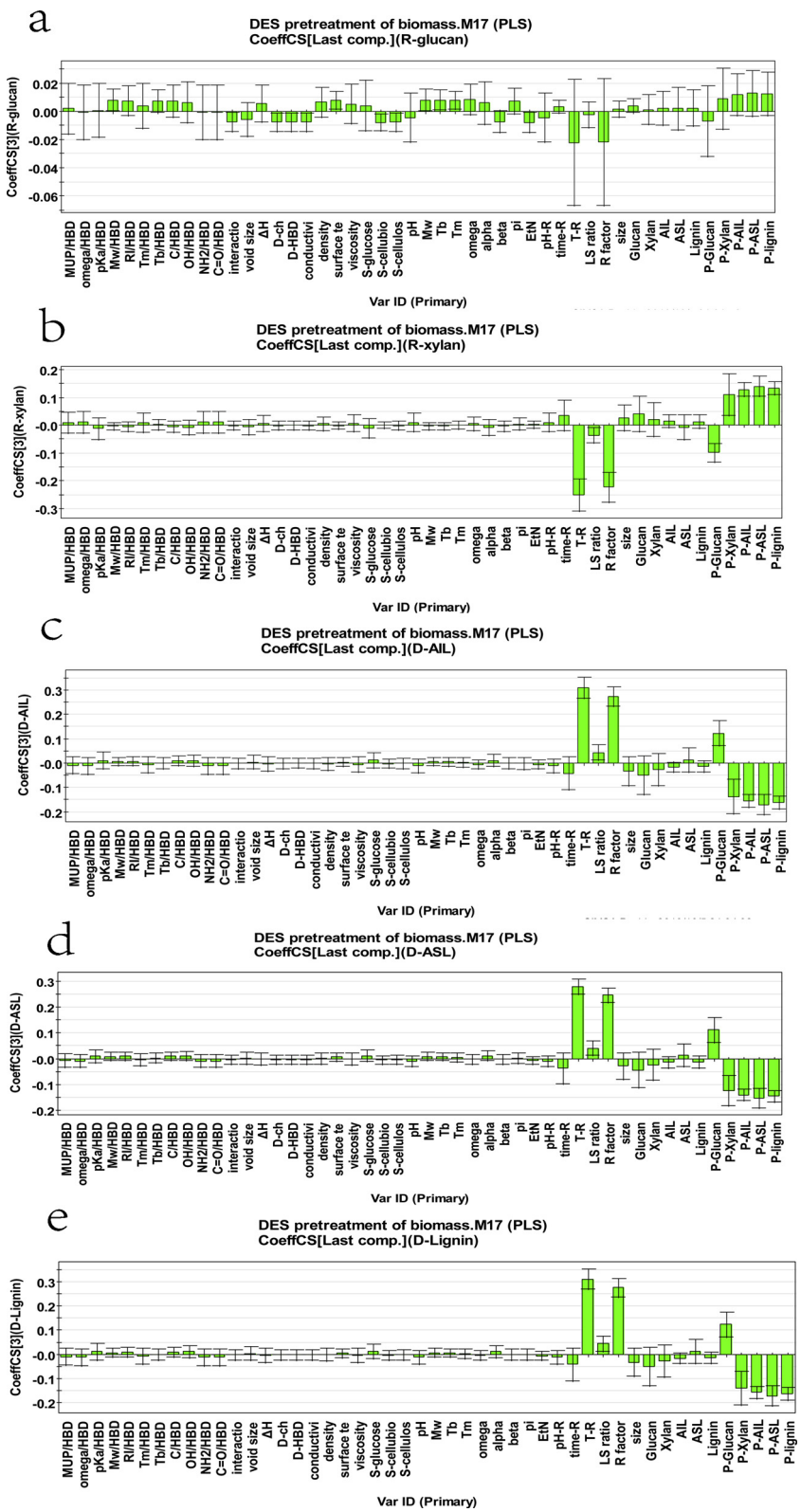


Fig. 6. The impact of all X variables on each Y variable. a) recovery rate of glucan as Y variable; b) recovery rate of xylan as Y variable; c) removal rate of acid insoluble lignin as Y variable; d) removal rate of acid soluble lignin as Y variable; e) removal rate of lignin as Y variable.

polarization parameter (pi) had strong relationship with glucan recovery rate, the acidity of DES (alpha) was closely related to the kpa of HBD. Self-diffusion coefficients of Choline chloride, self-diffusion coefficients of HBD and interaction energy of DES were located closely, which showed they had strong relationship between each other, for the reason that all these variables were on hydrogen bond network and spatial distribution functions of DES (Ferreira et al., 2020). As shown in Fig. 5c, VIP plot revealed the top five most important factors in this entire DES pretreatment dataset were as follows: temperature, severity of DES pretreatment, acid soluble lignin content of pretreated samples,

acid insoluble lignin content of pretreated samples.

Fig. 6 showed the impact of all 48 independent variables on each dependent variable. It was clear that with the increased of severity of DES pretreatment, the delignification rate increased and the recovery rate of carbohydrate decreased. The pretreatment temperature had positive relationship with D-lignin, D-ASL and D-AIL, and had negative relationship with R-glucan and R-xylan. Smaller particle size of raw materials and high ratio of biomass to DES would enhance the removal rate of lignin.

More hydroxyl structures in HBD could form more hydrogen bonds in DES, thus the OH/HBD had positive relationship with the recovery rate of glucan. DES with longer alkyl chain of HBD was more effective for removing lignin and recovering the glucan. Molecular weight, boiling point and melting point of HBD and DES had positive effect on enhancing the recovery of glucan and removal of lignin. Acidic DES showed better pretreatment in this discussion. The value of pka of HBD and alpha of DES had positive influence on the lignin removal. DES with lower viscosity had better mass transform ability and better delignification efficiency.

4. Conclusion

Multivariate analysis methods including principal component analysis and partial least squares were performed to investigate the inner relationships among variables of comprehensive DES pretreatment process. The effect of composition of biomass material, physicochemical properties of HBD and DES, pretreatment conditions on the DES pretreatment were discussed. All the analyzed models could explain most of the variability and had good predictive ability. The results revealed that pretreatment temperature, polarity, molecular weight, boiling point, mass transfer capacity, hydroxyl group content, acidity, and hydrogen bond strength were the key variables which showed strong positive correlation with removal of lignin and recovery of glucan. These observations had implications in designing of DES and optimizing of DES pretreatment for the biomass conversion.

CRediT authorship contribution statement

Huanfei Xu: Conceptualization, Methodology, Software, Data curation, Writing - original draft, Supervision, Writing - review & editing. **Yi Kong:** Data curation, Writing - original draft. **Jianjun Peng:** Data curation, Writing - original draft. **Xiaoming Song:** Investigation. **Xinpeng Che:** Data curation, Writing - original draft. **Shiwei Liu:** Software, Validation. **Wende Tian:** Writing - review & editing.

Declaration of Competing Interest

None.

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E-supplementary data of this work can be found in online version of the paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.indcrop.2020.112363.

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